Data Flow Diagram

Date	05 October 2026
Team ID	UNM-293 2026-2430
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Name	Kaviya sree.T
Maximum Marks	2 Marks

Work Assigned to Kaviya sree.T

Task: Design the Data Flow Diagram (DFD) for the FitBuddy system, showing how user fitness information moves through the system, how the Gemini model generates a personalized fitness plan, and how the generated plan is stored and delivered to the user.

FitBuddy DFD – Level 0 / Context Flow

External Entity	Process	Data Store	Data Flow
User	1.0 Collect User Profile & Fitness Goals	User Profile Database	Age/fitness level, goals, preferences, available equipment
User	2.0 Generate Personalized Fitness Plan	Fitness Plan Database	Personalized workout plan, exercise schedule, recommendations
Gemini Models	3.0 AI Plan Processing & Validation	—	Prompt/context → AI-generated plan → validated response
User	4.0 Display & Update Fitness Plan	Fitness Plan Database	View plan, progress updates, revised preferences

DFD Symbol / Flow Guide

Component	FitBuddy Representation
External Entity	User / Gemini Models
Process	Collect Profile → Generate Plan → Validate → Display/Update
Data Store	User Profile Database / Fitness Plan Database
Data Flow	Profile data, fitness goals, AI prompt, generated plan, progress updates

Note: The DFD structure follows the uploaded reference, which defines external entities, processes, data stores, and labeled data flows as the key DFD elements.